Spectral and Particle-Level Falsifications

Target: Electrons, protons, neutrinos, and internal algebra structure.

1. Dimension spectral D_s plateau
   - Measure deviations $from\ D_s \approx 2.97$ in high-precision quantum systems.
   - Falsification: If experiments detect D_s outside the predicted band, the spectral attractor is invalid.
2. Particle mass hierarchies
   - Compare predicted mass ratios (e.g., m_μ/m_e, m_t/m_c) derived from eigenvalues of the Laplacian.
   - Falsification: Observed ratios outside model-predicted ranges.
3. Number of fermion families
   - The model predicts exactly 3 stable families due to the minimal internal algebra.
   - Falsification: Detection of a fourth fundamental family would directly contradict the MPFS structure.
4. Charge quantization
   - Check whether the discrete charges emerge consistently from the internal algebra.
   - Falsification: Observation of fractionally charged free particles would violate the model.

Cosmological-Level Falsifications

Target: CMB, dark energy $w(z)$, large-scale structure.

1. CMB spectral signatures
   - The model predicts subtle correlations between $D_s$variations and primordial spectrum.
   - Falsification: If precise measurements of the CMB show no correlation with predicted D_s-dependent anisotropies.
2.
3. Dark energy equation of state
   - Model predicts w(z) variations correlated with neutrino mass sum Σm_ν.
   - Falsification: Any observation of w(z) evolution inconsistent with these correlations.
4. Large-scale structure / super voids
   - Predicted number and size distributions of super voids correspond to the spectral projection.
   - Falsification: Observed void statistics incompatible with $D_s \approx 2.97$ predictions.
5. Hubble tension
   - If MPFS provides a dynamical explanation, H_0 must lie within a narrow predicted range.
   - Falsification: Precision measurements outside this predicted range.

Black Hole and Horizon Tests

1. Entropy bounds
    - Model links minimal information loss to horizon entropy.
    - Falsification: Observation of entropy exceeding predicted bounds, e.g., black holes with unexpected microstates.
2. Evaporation rates
    - Hawking-like evaporation should follow the spectral constraints.
    - Falsification: Measured deviations from predicted evaporation rates.

Experimental Quantum Tests

1. Atomic interferometry
    - Sub-millimeter gravity experiments could detect deviations from standard Newtonian predictions due to spectral projection effects.
    - Falsification: No deviation observed within predicted sensitivity bounds.
2. Entanglement and decoherence
    - Predicts residual correlations due to incomplete information projection.
    - Falsification: Violation of predicted entanglement entropy patterns.

Summary

MPFS can be falsified at multiple scales:

- Microscopic: particle masses, family number, charge quantization, neutrino spectral effects.
- Cosmological: CMB correlations, $w(z)$, super void statistics, H_0 predictions.
- Black hole physics: entropy bounds, evaporation rates.
- Quantum tests: interferometry, decoherence patterns.

Any single strong violation of these predictions would challenge the internal consistency of the MPFS model.